cse

Project Title	Smart Health Symptom Checker
Student Name	Sanjana Yadav (correct?)
Reg. Number	25BCY10270 (as shared earlier)
Course / Branch	CSE
College Name	VIT Bhopal
Subject Name	

🩺 Project Topic Finalized

Smart Health Symptom Checker and Disease Prediction System

Project Overview / Abstract

This web-based application allows users to input various symptoms and receive probable disease predictions using machine learning models. The system also provides recommended precautionary measures and allows users to generate health reports. The project focuses on improving accessibility of preliminary medical knowledge and reducing overload on hospitals for basic diagnosis.

Major Functional Modules (3+)

Module	Description
User Module	Login/Signup, Profile, Symptoms input
Prediction Engine	ML-based disease prediction from symptoms

Report & Suggestion Module	Prevention tips, medicines suggestion
Admin Panel (optional advanced)	Manage dataset, monitor records

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Non-Functional Requirements (4+)

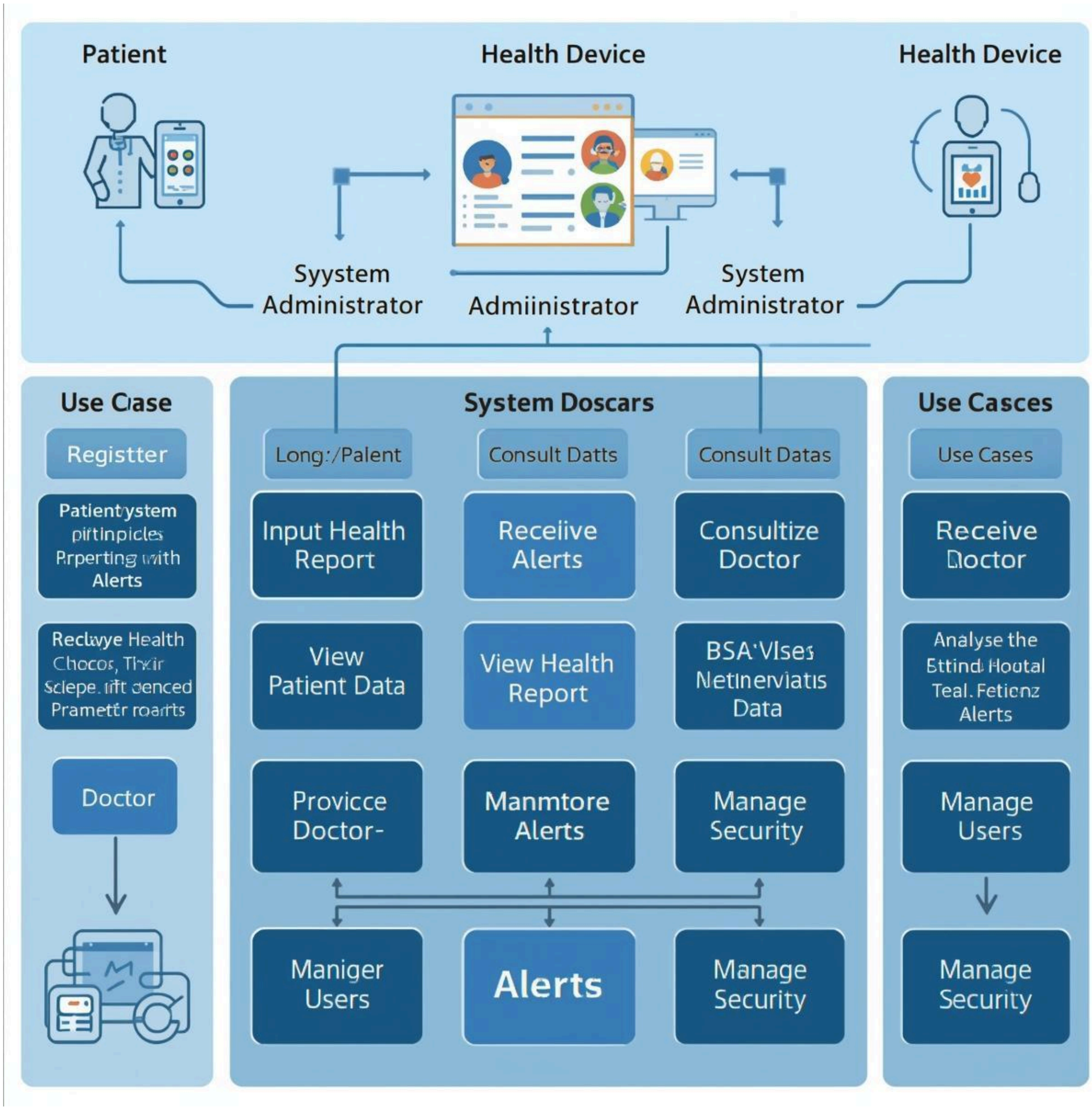
- Performance: Prediction response under 3 seconds
- Security: Hashing for passwords, role-based authentication
- Scalability: Support up to 10,000+ users with cloud database
- Usability: Clean UI with step-by-step symptom input
- Reliability: 98% uptime & data backup strategy
- Error handling: Input validation, exception logs

📊 Tech Stack

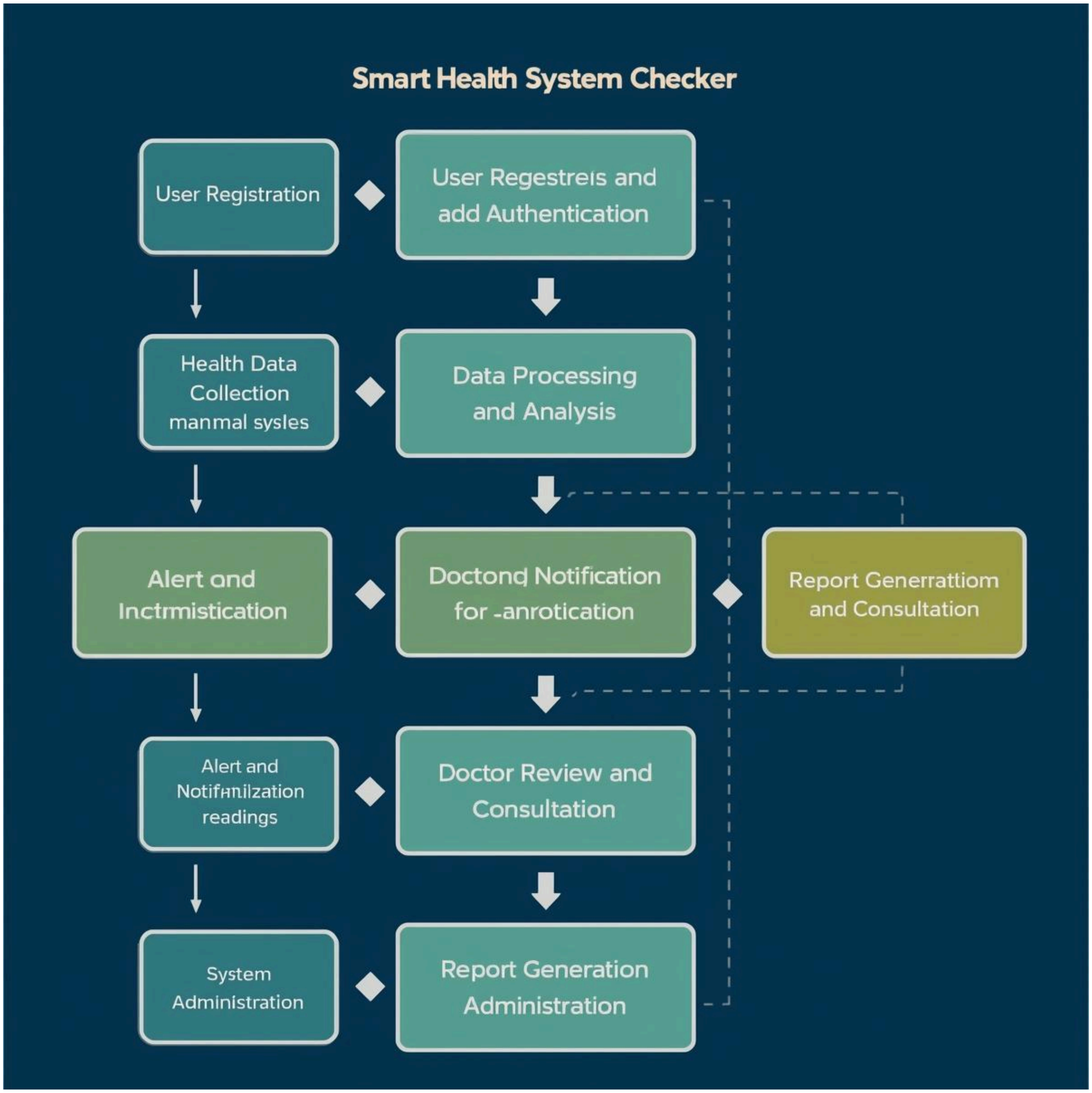
Layer	Tools / Technology
Frontend	HTML, CSS, JS, React (optional)
Backend	Python Flask / Django OR Node.js
Database	MongoDB / MySQL
ML Model	Naive Bayes / Random Forest / SVM
Hosting	GitHub + Render/Netlify

📎 UML + Diagram

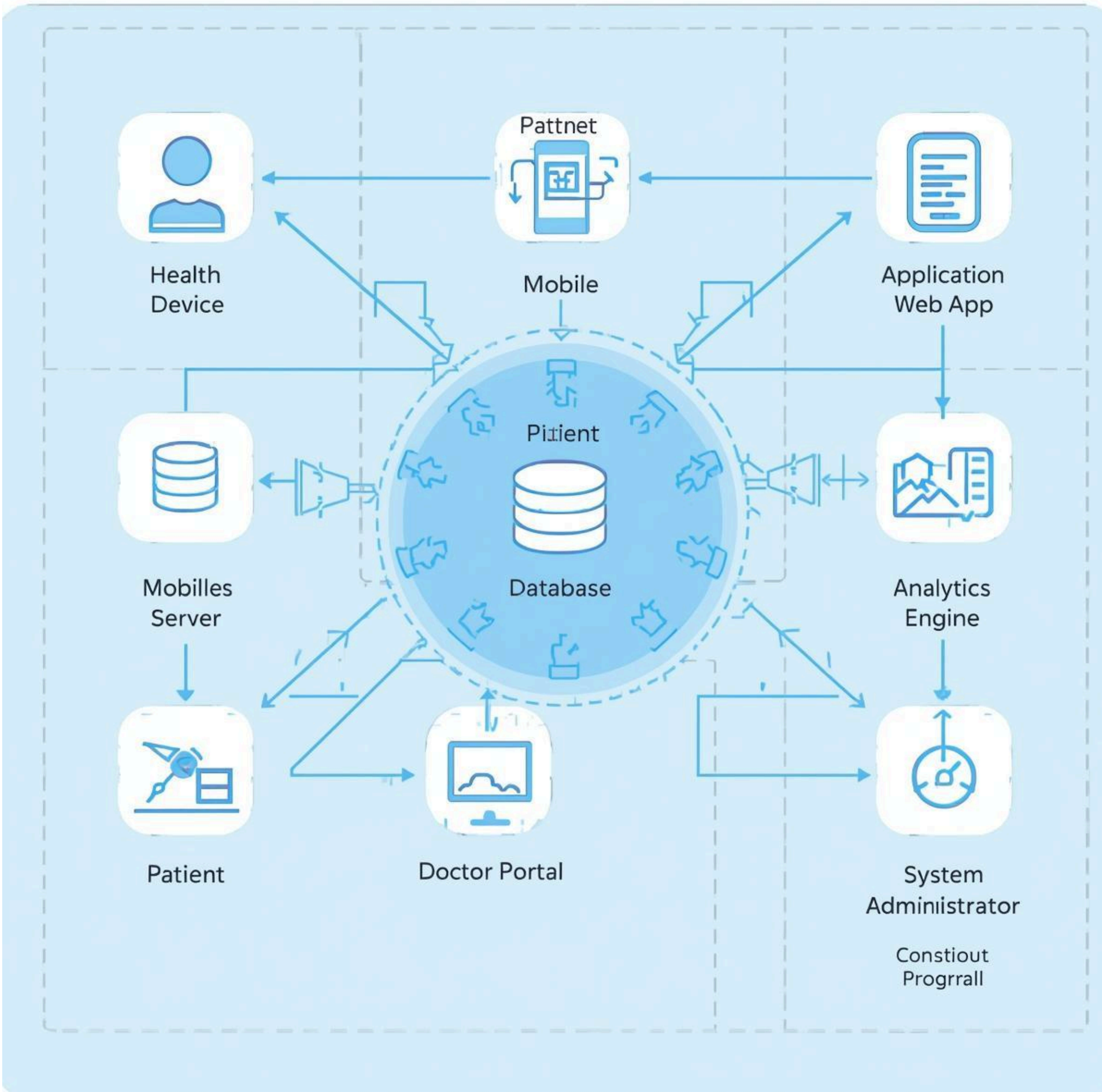
Use case diagram



- Workflow diagram



- System architecture



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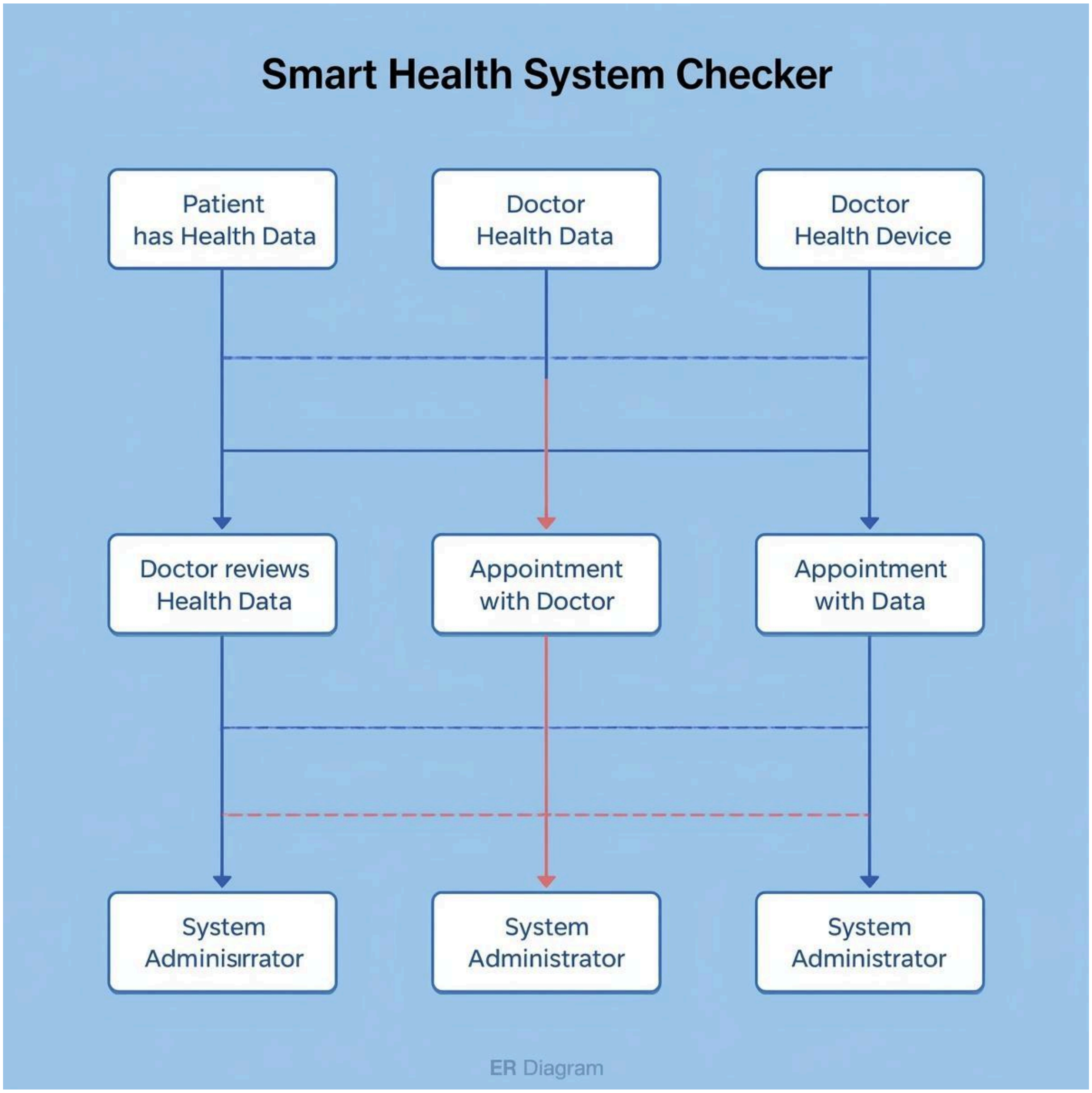
- ER Diagram



Future Enhancements

- Voice symptom detection
- Doctor live consultation
- Wearable health API integration


THANK
YOU

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